Quiz 04. Analysis of plane trusses
prepared by C. Cahig *for* ENS161 (AY 2026-2027, S1)

Instructions. Unless otherwise stated, all trusses are ideal. Take the acceleration due to gravity to be 9.80665. Express support reactions as Cartesian vectors. Express member forces as signed values: those in compression being negative[1].

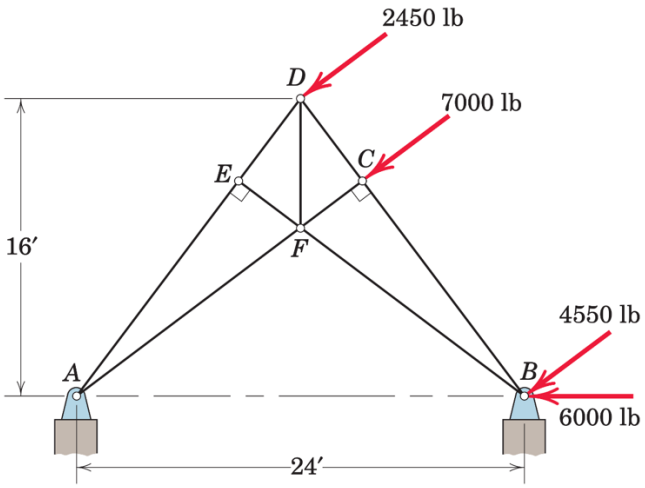


Figure 1. Roof truss for Scenario 1.

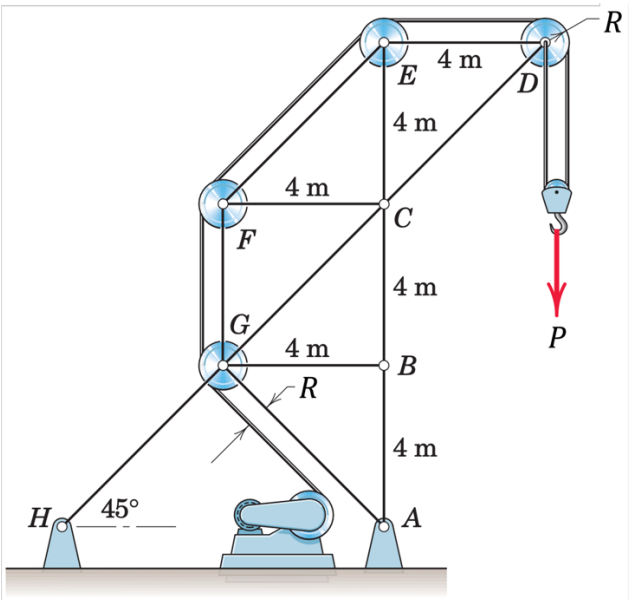


Figure 2. Truss with cable-and-pulley system for Scenario 2.

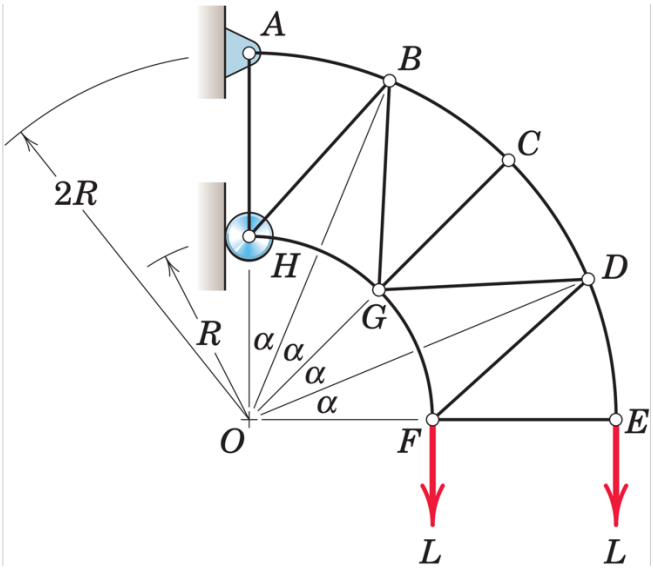


Figure 3. Arch truss for Scenario 3.

Scenario 1. A simplified roof truss model is supported on columns at A and B via pins. The loading simulates a case when the roof is barraged by such typhoon winds that render member DF broken. Determine:

a. the force in AE;
b. the force in AF;
c. the force in BF;
d. the force in BC;
e. the reaction at A; and
f. the reaction at B.

Scenario 2. With $R = 25$ cm and $P = 50$ kN, determine[2]:

a. the tension in the pulley cable;
b. the force in CD;
c. the force in FG;
d. the force in CG;
e. the force in BC;
f. the force in EF;
g. the force in GH;
h. the force in AG;
i. the reaction at H; and
j. the reaction at A.

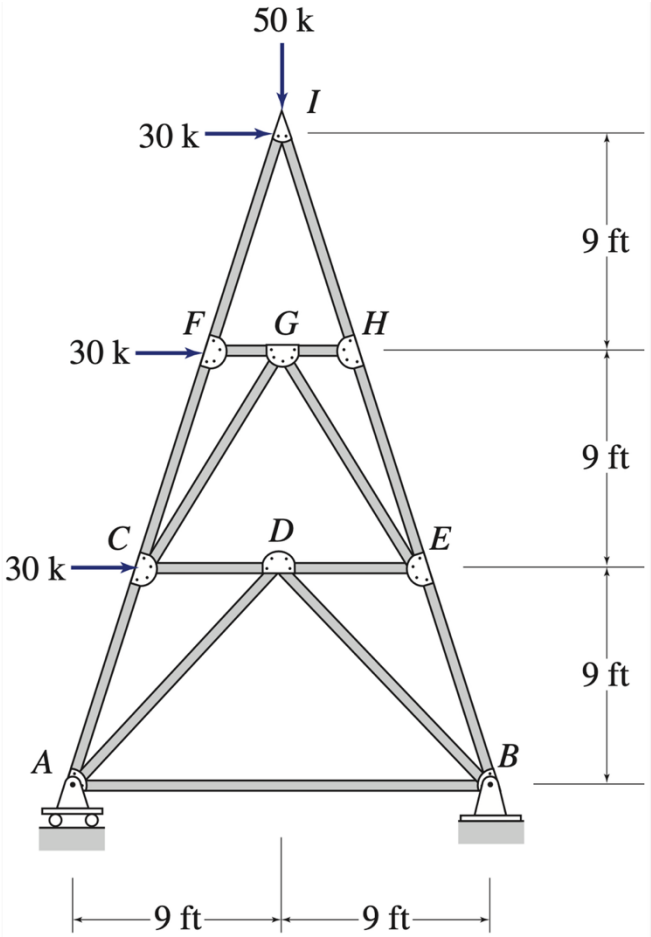
Scenario 3. The simple truss contains circular-arc members each subtending an angle of $\alpha = 22.5°$. Determine the forces[3]:

a. in DE;
b. in EF;
c. in GF;
d. in DF;
e. in DG; and
f. in CD

per unit value of L.

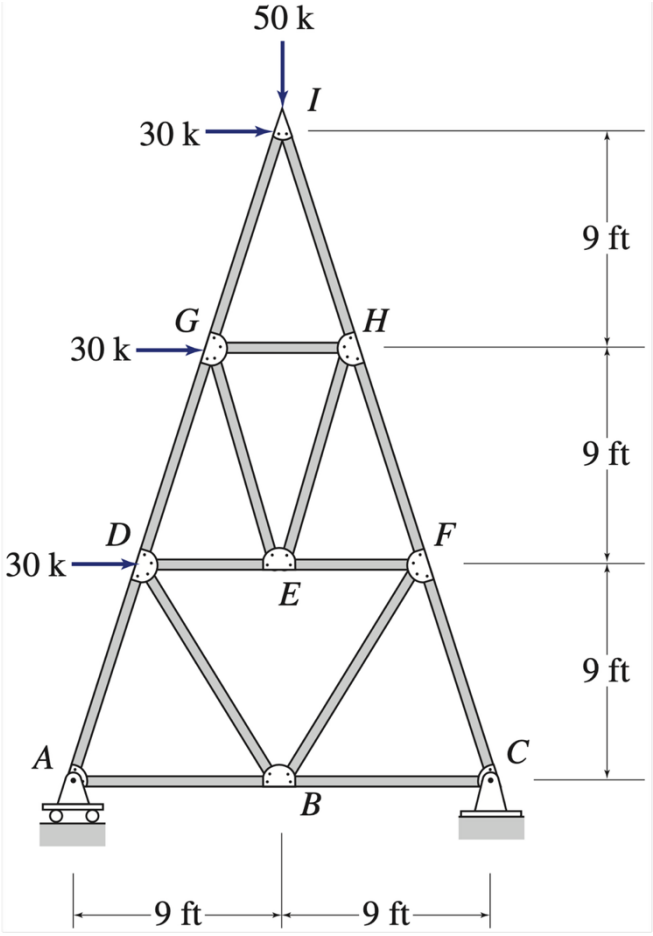
[1] For example, write -1.000 for a compressive newton and 1.000 for a tensile newton.
[2] At each pulley whose axle coincides with a joint, one may replace the cable forces with an equivalent force-couple system at the joint.
[3] The curved members are two-force members. Thus, the effect of each curved member is the same as that of a straight member.

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Scenario 4. For the given tower truss, determine:

a. the force in CF;
b. the force in CG;
c. the force in EG;
d. the force in EH;
e. the reaction at B; and
f. the reaction at A.



Scenario 5. For the given tower truss, determine:

a. the force in DG;
b. the force in EG;
c. the force in EH;
d. the force in FH;
e. the reaction at A; and
f. the reaction at C.

Marking. Each numbered item above is worth three (3) points. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.